Summary in Graph

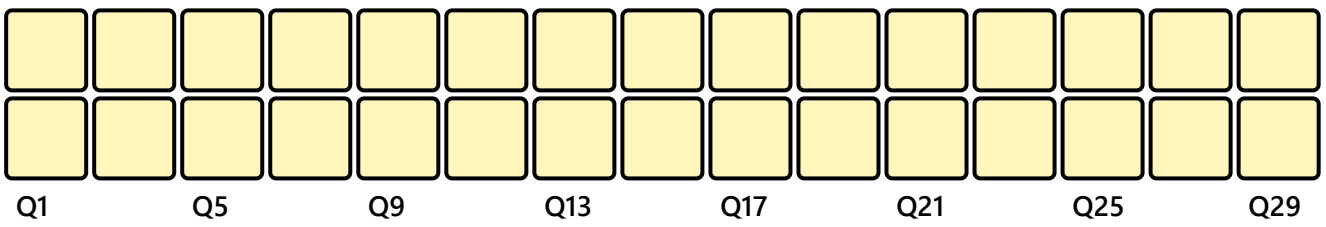
Exam Summary (GO Classes CS Test Series 2026 | DBMS | Subject Wise Test 3)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	30 (10 + 20)
Total Marks:	50 (10 + 40)
Exam Duration:	90 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (16) out of 158 users

EXAM RESPONSE EXAM STATS FEEDBACK

Technical

Q #1 Numerical Type Award: 1 Penalty: 0 Databases

Consider a B+ Tree of order $d = 2$ (Note: Order is defined as maximum pointers = $2d + 1$, minimum entries = d).

- Max pointers per node = 5.

- Max keys per node = 4.
- Min keys per node = 2.

We insert the following keys in this specific order into an initially empty tree:

$$1, 2, 3, 4, 5, 6, 7, 8, 9, 10$$

However, the insertion strategy is Standard Insertion (splitting nodes as they fill), NOT "Bulk Loading".

How many node splits (including leaf and internal nodes) occur during this entire process?

Your Answer: | Correct Answer: 3 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Select Type | Award: 1 | Penalty: 0 | Databases

In a database using WaitDie deadlock prevention protocol:

- Transaction T_{old} has timestamp 10.
- Transaction T_{young} has timestamp 20.

  (Smaller timestamp = Older transaction)

Which of the following scenarios are allowed/true?

A. If T_{old} requests a lock held by T_{young}, T_{old} waits.

B. If T_{old} requests a lock held by T_{young}, T_{old} is aborted (dies).

C. If T_{young} requests a lock held by T_{old}, T_{young} waits.

D. If T_{young} requests a lock held by T_{old}, T_{young} is aborted (dies).

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Numerical Type | Award: 1 | Penalty: 0 | Databases

Table T has attributes $(\mathrm{A}, \mathrm{B}, \mathrm{C})$. A is the Primary Key.

Query:

```
SELECT T1.A
FROM T AS T1, T AS T2
WHERE T1.C = T2.C
```

If table T has 1000 tuples, and the attribute C has 1000 distinct values (all unique), what is the cardinality of the result set?

Your Answer: | Correct Answer: 1000 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Databases

A database file has $1,000,000$ records.The file is ordered by the primary key ID.

- Block Size = 4096 bytes
- Record Size = 100 bytes
- Key Size = 10 bytes
- Pointer Size = 6 bytes

We construct a Primary Sparse Index on ID.
What is the maximum number of block I/Os required to fetch a random record using this index (worst case), assuming the index and data are on disk and no caching?

A. $\lceil \log_2(1000000) \rceil + 1$

B. $\lceil \log_{256}(25000) \rceil + 1$

C. $\lceil \log_{256}(25000) \rceil + 2$

D. $\lceil \log_{256}(1000000) \rceil + 1$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Numerical Type | Award: 1 | Penalty: 0 | Databases

Consider a relation schema

$$\mathrm{R(E, F, G, H)}$$

It is given that:

- E is a key for R.
- No other non-trivial functional dependencies are known to hold.
- Attribute E is atomic and does not contain NULL values.
- The term superkey is used in its formal relational sense.

The maximum number of superkeys for the relation schema $\mathrm{R(E, F, G, H)}$ is _____

Your Answer: | Correct Answer: 8 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Databases

Consider the following relational schemas used in a university library database:

$$\mathrm{Book(Title, Author, Catalog_{no}, Publisher, Year, Price)}$$

$$\mathrm{Collection(Title, Author, Catalog_{no})}$$

The following functional dependencies are known to hold:

1. $\{\mathrm{Title, Author}\} \rightarrow \mathrm{Catalog_{no}}$

2. $\text{Catalog}_{\text{no}} \rightarrow \text{Title}, \text{Author}, \text{Publisher}, \text{Year}$

3. $\{\text{Publisher}, \text{Title}, \text{Year}\} \rightarrow \text{Price}$

Assume that {Title, Author} is a candidate key for both Book and Collection.
Which of the following statements is TRUE?

A. Both Book and Collection are in BCNF

B. Both Book and Collection are in 3NF only

C. Book is in 2NF but not in 3NF, and Collection is in 3NF

D. Both Book and Collection are in 2NF only

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Databases

Consider a relation schema

$$\text{R}(\text{A}, \text{B}, \text{C}, \text{D}, \text{E})$$

With the following set of functional dependencies:

$$F = \{\text{A} \rightarrow \text{B}, \text{C} \rightarrow \text{D}, \text{AC} \rightarrow \text{E}\}$$

A student proposes the following decomposition of R :

$$\text{R1}(\text{ A, B}), \text{R2}(\text{C}, \text{D}), \text{R3}(\text{ A}, \text{C}, \text{E})$$

Which of the following statements about the above decomposition is CORRECT?

A. The decomposition is lossless join and dependency preserving.

B. The decomposition is lossless join but not dependency preserving.

C. The decomposition is dependency preserving but not lossless join.

D. The decomposition is neither lossless join nor dependency preserving.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Select Type | Award: 1 | Penalty: 0 | Databases

Let A and B be two relations, and let C be an arbitrary join condition involving attributes from both A and B .

In relational algebra, the inner join of A and B with respect to condition C is denoted as:

$$A \bowtie_C B$$

Assume that:

- No outer join semantics are implied.
- No projection is applied unless explicitly stated, and
- Tuples are combined strictly according to relational algebra definitions.

Which of the following most accurately describes the result of the operation $A \bowtie_C B$?

A. For every pair of tuples (a, b) such that $a \in A$ and $b \in B$, the tuples are concatenated if and only if condition C evaluates to true; all other tuples are excluded from the result.

B. For every tuple in A, if no tuple in B satisfies condition C, the tuple from A appears in the result with NULL values for attributes of B.

C. For every tuple in B, if no tuple in A satisfies condition C, the tuple from B appears in the result with NULL values for attributes of A.

D. The result consists of all possible concatenations of tuples from A and B, irrespective of condition C.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Select Type | Award: 1 | Penalty: 0 | Databases

Let the relation schema be:

$$\mathrm{R(cid, title, year, dept)}$$

Assume the following set of functional dependencies holds on R :

$$\mathrm{F} = \{\mathrm{cid} \rightarrow \mathrm{title}, \mathrm{title} \rightarrow \mathrm{dept}, (\mathrm{cid}, \mathrm{year}) \rightarrow \mathrm{dept}\}$$

No other functional dependencies hold except those implied by Armstrong's axioms.
Which of the following tuple sets can exist in a legal instance of R ?

A. $(\mathrm{c1}, \mathrm{DB}, 2010, \mathrm{CS}), (\mathrm{c1}, \mathrm{DB}, 2011, \mathrm{EE})$

B. $(\mathrm{c1}, \mathrm{DB}, 2010, \mathrm{CS}), (\mathrm{c2}, \mathrm{DB}, 2012, \mathrm{CS})$

C. $(\mathrm{c1}, \mathrm{DB}, 2010, \mathrm{CS}), (\mathrm{c1}, \mathrm{AI}, 2011, \mathrm{CS})$

D. $(\mathrm{c1}, \mathrm{DB}, 2010, \mathrm{CS}), (\mathrm{c2}, \mathrm{AI}, 2012, \mathrm{EE})$

Your Answer: | Correct Answer: B;D | Not Attempted | Time taken: 00min 06sec | Discuss

Q #10 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Databases

Let $\mathrm{R(A, B, C, D)}$ be a relation schema, and let $X \rightarrow Y$ be a functional dependency defined on R.

1. A functional dependency is said to be trivial if $Y \subseteq X$.

2. A functional dependency is said to be completely non-trivial if no attribute on the right-hand side appears on the left-hand side.

Which of the following statements best characterizes a completely non-trivial functional dependency?

A. A functional dependency $\mathrm{X} \rightarrow \mathrm{Y}$ where $\mathrm{Y} \subseteq \mathrm{X}$

B. A functional dependency $\mathrm{X} \rightarrow \mathrm{Y}$ where $\mathrm{X} \cap \mathrm{Y} = \varnothing$

C. A functional dependency $\mathrm{X} \rightarrow \mathrm{Y}$ where $\mathrm{Y} \not\subset \mathrm{X}$

D. None of the above

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Numerical Type | Award: 2 | Penalty: 0 | Databases

Consider the following two relations $\mathbf{T1}(\mathbf{P}, \mathbf{Q})$ and $\mathbf{T2}(\mathbf{R}, \mathbf{S})$ with the instances shown below.

- In relation T1, attribute $\mathbf{p}$ is the primary key.
- In relation T2, attribute $\mathbf{R}$ is the primary key.
- Attribute $\mathbf{S}$ in T2 is a foreign key that references T1(P).

The referential integrity constraints are defined as follows:

- ON DELETE CASCADE
- ON UPDATE SET NULL

The current contents of the relations are:

$\mathrm{T1}(P, Q) =$

P	Q
2	2
3	8
5	8
6	9
7	3
8	5
9	8

$\mathrm{T2}(R, S) =$

R	S
2	2
3	2
5	7
7	2
8	3
9	7

Now, the following sequence of SQL operations is issued in the given order:

```
1. UPDATE T1 SET P = 10 WHERE P = 2;

2. DELETE FROM T1 WHERE P = 7;

3. UPDATE T1 SET P = 3 WHERE P = 8;
```

Assume that:

- All constraints are immediately enforced
- No other constraints exist
- No transaction rollback occurs unless explicitly forced by a constraint violation

After the execution of all the above operations, how many tuples remain in relation T2?

Your Answer: | Correct Answer: 4 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

Let $\mathrm{R}_1(\mathrm{a}, \mathrm{b}, \mathrm{c})$ and $\mathrm{R}_2(\mathrm{x}, \mathrm{y}, \mathrm{z})$ be two relations in a relational database, and the attribute "a" of relation R_1 is a foreign key that references the primary key x of relation R_2.

Assume that:

- No ON DELETE or ON UPDATE actions (CASCADE / SET NULL / SET DEFAULT) are specified.
- Referential integrity constraints are immediately enforced.
- All attributes are NOT NULL, unless forced otherwise by constraint semantics.

Consider the following operations issued independently (each operation is applied to the initial database state, not cumulatively):

1. Insert a tuple into R_1
2. Insert a tuple into R_2
3. Delete a tuple from R_1
4. Delete a tuple from R_2

Which of the following pairs of operations may cause a violation of the referential integrity constraint, depending on the data values involved?

A. Operations 1 and 2

B. Operations 2 and 3

C. Operations 3 and 4

D. Operations 4 and 1

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Select Type | Award: 2 | Penalty: 0 | Databases

Let the relation schema be:

$$\mathrm{R}(\mathrm{cid}, \mathrm{title}, \mathrm{year}, \mathrm{dept})$$

Assume the following set of functional dependencies holds on R:

$$\mathrm{F} = \{\mathrm{cid} \rightarrow \mathrm{title}, \mathrm{title} \rightarrow \mathrm{dept}, (\mathrm{cid}, \mathrm{year}) \rightarrow \mathrm{dept}\}$$

No other functional dependencies hold except those implied by Armstrong's axioms.
Which of the following statements are necessarily true?

A. $\{\mathrm{cid}\}$ is a candidate key of R

B. $\{\mathrm{cid}, \mathrm{year}\}$ is a superkey but not a candidate key

C. The dependency (cid, year) dept is redundant

D. Two tuples with the same title but different dept values can never co-exist in any legal instance of R

Your Answer: | Correct Answer: C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

Consider the following database schema:

- $\text{Sailors}(\text{sid}, \text{sname}, \text{rating}, \text{age})$
- $\text{Boats}(\text{bid}, \text{bname}, \text{color})$
- $\text{Reserves}(\text{sid}, \text{bid}, \text{day})$

Assume the following:

1. sid is the primary key of Sailors.
2. bid is the primary key of Boats.
3. (sid, bid, day) is the primary key of Reserves.
4. SQL follows set semantics (that is, duplicate tuples are eliminated).
5. Every sailor may reserve the same boat on multiple days.
6. Boats can have colors other than 'red' and 'green'.

Now consider the following SQL query:

```
SELECT S.sid
FROM Sailors S, Reserves R, Boats B
WHERE S.sid = R.sid
    AND R.bid = B.bid
    AND B.color = 'red'
EXCEPT
SELECT S2.sid
FROM Sailors S2, Reserves R2, Boats B2
WHERE S2.sid = R2.sid
    AND R2.bid = B2.bid
    AND B2.color = 'green';
```

Which of the following statements MOST ACCURATELY describes the result produced by the above query?

A. The query returns the sid of every sailor who has reserved at least one red boat and has never reserved any green boat.

B. The query returns the sid of every sailor who has reserved only red boats and no boat of any other color.

C. The query returns the sid of every sailor who has reserved all red boats in the database but no green boat.

D. The query returns the sid of every sailor who has reserved a red boat and a green boat on different days.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

Consider a relational table STUDENT, which stores exactly one tuple for each registered student in a university.
The table has the following attributes:

1. Registration_Num: A unique registration number assigned by the university to each registered student.

2. UID: A unique identity number assigned at the national level to each citizen.

3. BankAccount_Num: A bank account number.

- A student may have multiple bank accounts.
- A bank account may be a joint account shared by multiple individuals.
- This attribute stores only one (primary) account number for the student.

4. Name: Name of the student.

5. Hostel_Room: Hostel room number assigned to the student (if any).

Assume the following:

- The table contains no duplicate tuples.
- NULL values may appear wherever logically applicable.
- No additional integrity constraints are enforced other than those explicitly stated above.

Which of the following statements is INCORRECT?

A. BankAccount_Num is a candidate key.

B. Registration_Num can be chosen as a primary key.

C. UID is a candidate key if all students are from the same country.

D. If $\mathbf{S}$ is a superkey such that $\mathrm{S} \cap \mathrm{UID} = \varnothing$, then $\mathrm{S} \cup \mathrm{UID}$ is also a superkey.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

Consider a relation $\mathrm{R}(\mathrm{A}, \mathrm{B})$ where:

- Attribute A may contain duplicate values,
- Attribute B may contain duplicate values as well as NULLs.

Assume that the SQL engine strictly follows the SQL standard.
Let the following SQL queries be executed on relation R:

```
    Q1: SELECT COUNT(*) FROM R;

    Q2: SELECT COUNT(DISTINCT A) FROM R;

5.  Q3: SELECT COUNT(DISTINCT B) FROM R;

    Q4: SELECT COUNT(DISTINCT *) FROM R;

    Q5: SELECT MAX(DISTINCT A) FROM R;
```

Which of the following statements is CORRECT?

A. Queries Q1, Q2, Q3, and Q5 are valid SQL queries, but Q4 is syntactically invalid.

B. Queries Q2, Q3, and Q5 are valid SQL queries, but Q1 and Q4 are syntactically invalid.

C. Queries Q1, Q2, Q3, and Q4 are valid SQL queries, but Q5 is syntactically invalid.

D. Only queries Q1, Q2, and Q3 are valid SQL queries.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

Consider a relation $R(A, B)$ with the following multiset (bag) of tuples:

$$(1, 10)$$
$$(2, \text{NULL})$$
$$(3, \text{NULL})$$
$$(4, 10)$$
$$(5, \text{NULL})$$

Assume that the SQL system strictly follows the SQL standard regarding NULL values and aggregate function semantics.
Now consider the following SQL queries executed independently on relation R:

```
Q1: SELECT COUNT(*) FROM R;

Q2: SELECT COUNT(B) FROM R;

Q3: SELECT COUNT(DISTINCT B) FROM R;

Q4: SELECT SUM(B) FROM R;

Q5: SELECT AVG(B) FROM R;

Q6: SELECT SUM(DISTINCT B) FROM R;
```

Which of the following options gives the CORRECT ordered result

$$(\text{Q1}, \text{Q2}, \text{Q3}, \text{Q4}, \text{Q5}, \text{Q6})?$$

A. $(5, 2, 1, 20, 10, 10)$

B. $(5, 2, 1, 20, 20, 10)$

C. ($5, 5, 2$, NULL, NULL, NULL)

D. $(5, 2, 2, 20, 10, 20)$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 | Multiple Select Type | Award: 2 | Penalty: 0 | Databases

Consider a relation $\mathrm{R}(\mathrm{A}, \mathrm{B}, \mathrm{C})$, where C is an integer-valued attribute that is allowed to take NULL values. Assume that the SQL system follows the SQL standard semantics, including three-valued logic (TRUE, FALSE, UNKNOWN).
Consider the following SQL query:

```
SELECT *
FROM R
WHERE (C>10)OR(C<=10);
```

Which of the following statements about the result of the above query are CORRECT?
(Select ALL correct options.)

A. Every tuple in R for which C is NOT NULL will appear in the result.

B. Every tuple in R for which C is NULL will appear in the result.

C. The result of the query is identical to the result of: `SELECT * FROM R WHERE C IS NOT NULL;`

D. For any tuple t in R, if C evaluates to UNKNOWN in both comparisons, then t is excluded from the result.

Your Answer: | Correct Answer: A;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 | Multiple Select Type | Award: 2 | Penalty: 0 | Databases

Consider the following relation:

$$\mathrm{Scores}(\mathrm{Team}, \mathrm{Day}, \mathrm{Opponent}, \mathrm{Runs})$$

Given that:

- Each tuple represents the number of runs scored by Team against Opponent on a given Day.
- A team plays at most one match per day.
- The attribute Runs is an integer and never NULL.
- There are multiple matches on the same day.

Query Q1

```
SELECT S1.Team, S1.Day
FROM Scores S1
WHERE S1.Runs <= ALL (
        SELECT S2.Runs
        FROM Scores S2
        WHERE S2.Day = S1.Day
    );
```

Query Q2

```
SELECT S1.Team, S1.Day
FROM Scores S1
WHERE NOT EXISTS (
        SELECT *
        FROM Scores S2
        WHERE S2.Day = S1.Day
            AND S2.Runs < S1.Runs
    );
```

Query Q3

```
SELECT S1.Team, S1.Day
FROM Scores S1
WHERE S1.Runs = (
        SELECT MIN(S2.Runs)
        FROM Scores S2
        WHERE S2.Day = S1.Day
    );
```

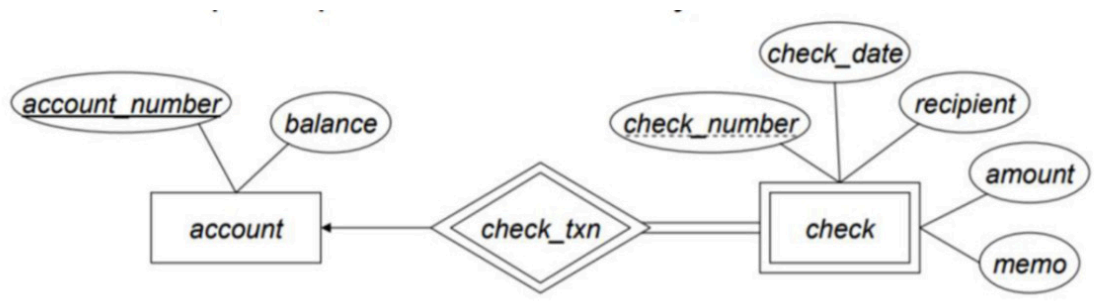
Which of the following statements are CORRECT?

A. Query Q1 and Query Q2 always produce identical results.
B. Query Q1 and Query Q3 always produce identical results.
C. Query Q2 and Query Q3 always produce identical results.
D. For a given day, if multiple teams have the same minimum runs, all such teams appear in the output of all three queries.

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 | Multiple Select Type | Award: 2 | Penalty: 0 | Databases

Consider the following Entity-Relationship (ER) model of a banking system.



There are two entity sets:

1. Account, with attributes: account_number (key attribute), balance

2. Check, with attributes: check_number (partial key), check_date, recipient, amount, memo

The entity set Check is related to Account through a relationship check_txn, as shown in the ER diagram. The relationship check_txn is an identifying relationship, and Check is a weak entity set.
Assume the following semantics derived from the ER diagram:

- Every Check is associated with exactly one Account.
- An Account may be associated with zero or more Checks.
- A Check cannot exist without being associated with an Account.
- The attribute check_number is unique only within an Account, but not globally unique.
- No additional constraints are assumed unless logically implied by the ER model.

Which of the following statements is/are CORRECT?

A. The primary key of the relation corresponding to Check must include account_number along with check_number.

B. The participation of Check in the relationship check_txn is total, whereas the participation of Account in check_txn is partial.

C. The relationship check_txn must be many-to-one from Check to Account.

D. It is possible for two different Accounts to have Checks with the same check_number.

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 | Multiple Select Type | Award: 2 | Penalty: 0 | Databases

Consider the following ER model used by a financial institution.

- Customer is a strong entity with attributes: cust_id (primary key), name, and city.

- Loan is a strong entity with attributes: loan_id (primary key) and amount.
- Payment is a weak entity with attributes: payment_no (partial key) and payment_date.
- A Payment is related to Loan via an identifying relationship pays_for.

The model satisfies the following constraints:

1. Every Payment is associated with exactly one Loan.
2. A Loan may have zero or more Payments.
3. Payment has total participation in pays_for.
4. payment_no is unique only within a Loan, not across all Loans.
5. No other business rules are assumed.

The ER model is converted into relations using standard ER-to-relational mapping rules.
Which of the following statements are TRUE? (Select ALL that apply)

A. The primary key of the relation corresponding to Payment must be (loan_id, payment_no).

B. Deleting a Loan tuple may require deletion of multiple Payment tuples.

C. It is possible for two different Loans to have a Payment with the same payment_no.

D. All constraints of the ER model can be enforced using only PRIMARY KEY, FOREIGN KEY, and NOT NULL constraints in SQL.

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #22 | Multiple Select Type | Award: 2 | Penalty: 0 | Databases

Consider a database with two integer variables A and B.
The only consistency constraint on the database is:

$$0 <= \mathrm{A} <= \mathrm{B}$$

A transaction is said to preserve consistency if for every initial state satisfying the constraint, the final committed state also satisfies the constraint.
Given:

- Each transaction executes atomically
- Statements inside a transaction are executed sequentially
- No overflow occurs
- No concurrency (single transaction at a time)

Transaction T:

$$\begin{aligned} &1.\ A := A + B \\ &2.\ B := B + 1 \\ &3.\ A := B - A \\ &4.\ B := A + B \end{aligned}$$

Which of the following statements are TRUE?

A. Transaction T preserves the consistency constraint for all valid initial states.

B. Transaction T violates the constraint for some valid initial state.

C. At some intermediate step, the constraint may be violated, but the final state always satisfies the constraint.

D. There exists an initial state satisfying the constraint such that after step (1) itself, the constraint is violated.

Your Answer: | Correct Answer: B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #23 | Multiple Select Type | Award: 2 | Penalty: 0 | Databases

Consider a database system that supports transactions with the following guarantees:

- Atomicity is implemented using write-ahead logging (WAL).
- Durability is ensured by forcing the log to stable storage at commit.
- Isolation is implemented using strict two-phase locking (Strict 2PL).
- Consistency is enforced only through application-level constraints, not by the DBMS itself.

Assume the database enforces the invariant $A + B = 100$, where A and B are integer-valued database items.

Two transactions T_1 and T_2 execute concurrently as follows:

Transaction T_1:

```
read(A);
A := A - 10;
write(A);
Commit;
```

Transaction T_2:

```
read(B);
B := B + 10;
write(B);
Commit;
```

Assume:

- Both transactions individually preserve the invariant when executed alone.
- A system crash occurs immediately after both transactions have written their updates to the database buffer but before any commit record is written to disk.
- Recovery follows the standard ARIES protocol.

Which of the following statements is/are TRUE?

A. Atomicity guarantees that either both updates of each transaction appear in the database or none appear.

B. After recovery, the database state is guaranteed to satisfy the invariant $\mathrm{A} + \mathrm{B} = 100$.

C. Durability plays no role in preserving the invariant in this execution.

D. It is possible for the database to violate consistency after recovery even though Atomicity, Isolation, and Durability are all correctly implemented.

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #24 | Numerical Type | Award: 2 | Penalty: 0 | Databases

Consider two database items $\mathbf{A}$ and $\mathbf{B}$, with initial values:

$$A = 25, B = 25$$

Two transactions T_1 and T_2 execute concurrently under an unspecified concurrency control protocol, following the interleaved schedule shown below.
Transaction T_1

```
   READ(A, t)
   t := t + 100
   WRITE(A, t)
   READ(B, t)
5. t := t + 100
   WRITE(B, t)
```

Transaction T_2

```
   READ(A, s)
   s := s * 2
   WRITE(A, s)
   READ(B, s)
5. s := s * 2
   WRITE(B, s)
```

Assume:

- All read and write operations occur exactly in the order shown.
- No transaction aborts.
- There is no crash.
- The DBMS does not enforce serializability.
- All writes overwrite the previous value immediately.

Let $S = A + B$ be the final sum of database values after both transactions are complete.

What is the value of S ?

Your Answer: | Correct Answer: 500 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

Consider the following schedule S consisting of three transactions T_1, T_2, and T_3 operating on data items X, Y, and Z.
(Note: $R(A)$ denotes Read A, and $W(A)$ denotes Write A).

$$S : R_1(X), W_2(X), W_1(X), R_3(X), W_3(Y), W_2(Y), C_1, C_3, C_2$$

Which of the following statements is TRUE regarding schedule S?

A. S is Conflict Serializable but not Recoverable.

B. S is View Serializable but not Conflict Serializable.

C. S is not View Serializable because of a blind write on X.

D. S is Recoverable and Cascadeless.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #26 | Numerical Type | Award: 2 | Penalty: 0 | Databases

Consider a $\mathrm{B}+$ Tree with the following specifications:

- Block Size: 1024 Bytes
- Block Pointer Size: 8 Bytes
- Record Pointer Size: 9 Bytes
- Search Key Size: 15 Bytes

The nodes are completely filled to their maximum capacity (100% fill factor for calculation). If the B+ tree is used to index a table with $1,000,000$ (1 million) records, what is the minimum depth (height) of the index? (Assume root is at level 1, children at level 2, etc.)

Your Answer: | Correct Answer: 4 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 | Multiple Select Type | Award: 2 | Penalty: 0 | Databases

Let $R(A, B, C, D, E)$ be a relation schema with the following Functional Dependencies (FDs):

$$F = \{AB \rightarrow C, \quad C \rightarrow D, \quad D \rightarrow E, \quad E \rightarrow A\}$$

Which of the following statements are CORRECT?

A. The decomposition of R into $R_1(A, B, C)$ and $R_2(C, D, E)$ is Lossless.

B. The relation R is in 3NF but not in BCNF.

C. The attribute B is a prime attribute.

D. The decomposition of R into $R_1(A, B, C)$, $R_2(C, D)$, $R_3(D, E)$, $R_4(E, A)$ is Dependency Preserving.

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #28 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

Consider the following schedule involving 3 transactions:

$$S : R_1(X), R_2(Z), R_1(Z), R_3(X), R_3(Y), W_1(X), C_1, W_3(Y), C_3, R_2(Y), W_2(Z), W_2(Y), C_2$$

Why does the Thomas Write Rule fail to make this schedule Serializable, or does it?

A. Thomas Write Rule applies, making it Serializable.

B. Thomas Write Rule only applies to obsolete writes; here we have Read-Write conflicts that form a cycle.

C. The schedule is not View Serializable.

D. Both B and C are correct.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

Table **Emp** has column **Salary** with values: $\{1000, 2000, \text{NULL}, 3000\}$.

Consider the query:

```
SELECT count(*) FROM Emp
WHERE Salary > ALL (SELECT Salary FROM Emp WHERE Salary = 2000);
```

What is the output?

A. 1 (The value 3000)

B. 2 (Values NULL and 3000)

C. 0

D. Error

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

Consider a B+ Tree index constructed on a database relation containing $\text{N} = 1,000,000$ (one million) records.
The system parameters are as follows:

- Disk Block Size: 1024 Bytes
- Search Key Size: 14 Bytes
- Block Pointer Size (Tree Pointer): 8 Bytes
- Record Pointer Size (Data Pointer): 10 Bytes
- Leaf Node Structure: Each leaf node contains a sequence of $(\text{Key}, \text{RecordPointer})$ pairs and one "Next Leaf" pointer (8 Bytes) to maintain the linked list.
- Internal Node Structure: Each internal node contains a sequence of K keys and $\text{K} + 1$ block pointers.

Assume the B+ Tree is in its worst-case space utilization (i.e., every node is filled to the minimum requirement allowed by the B+ Tree properties).

What is the maximum height of this B+ Tree?
(Note: Root is at Height 1).

A. 4

B. 5

C. 6

D. 7

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

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